



Statistics of Financial Markets

Chapter 8: Volatility estimators



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Acronyms used in this material

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARCH-LM: ARCH Lagrange Multiplier (test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CBOE: Chicago Board Options Exchange
- ▣ DM: Diebold–Mariano (test)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GK: Garman–Klass (volatility estimator)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OHLC: Open, High, Low, Close
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- ▣ VIX: CBOE Volatility Index
- ▣ YZ: Yang–Zhang (volatility estimator)

Today's Question and Route

- **Question:** how large is the risk of an asset today, if we can never observe it directly?
 - ▶ Chapter 7: returns are almost unpredictable, but their size is not; here we measure that size
 - ▶ Chapter 9 turns the measurements into a model (ARCH and GARCH)
- **Route** of the chapter
 - ▶ what volatility is and why it is latent; historical volatility and the choice of the window
 - ▶ EWMA (RiskMetrics); range-based estimators from open, high, low and close prices
 - ▶ realised volatility; volatility clustering and its tests
 - ▶ long-run behaviour, the VIX, the leverage effect; how to compare volatility forecasts

Learning Outcomes

- ▣ Define volatility, annualise it with the actual trading frequency and explain why it is a latent quantity
- ▣ Compute historical and EWMA volatility and choose a window or a decay factor
- ▣ Compute the Parkinson, Garman–Klass, Rogers–Satchell and Yang–Zhang estimators and state their assumptions
- ▣ Detect volatility clustering with the ACF of squared returns, the Ljung–Box test and the ARCH-LM test
- ▣ Compare implied and realised volatility, and evaluate volatility forecasts with MSE and QLIKE

Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 13 (introduction: volatility and conditional heteroskedasticity)
 - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 13
- Volatility and its stylised facts: [Tsay \(2010\)](#), Ch. 3.1; a review of volatility forecasting: [Poon & Granger \(2003\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_08](#)
 - ▶ ported from the SFE Quantlets SFEkurgarch (kurtosis of a GARCH process) and SFEvolnonparest (nonparametric volatility)
- Lecture notebook: [open in Google Colab](#)
- Video course: [!\[\]\(2824aab9645d9fab95bae27ff6828dab_img.jpg\) Applied Time Series Analysis with Python](#)

Volatility Comes in Bursts

- “Large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes” [Mandelbrot \(1963\)](#)
 - ▶ after the crash of 19 October 1987, markets stayed turbulent for weeks
- Volatility enters almost every number in finance
 - ▶ risk measures: VaR and ES ([Chapter 10](#)), the Sharpe ratio ([Chapter 1](#))
 - ▶ option prices, margins of exchanges, the size of positions
- This chapter: how to **measure** it from prices, and how to tell whether it clusters



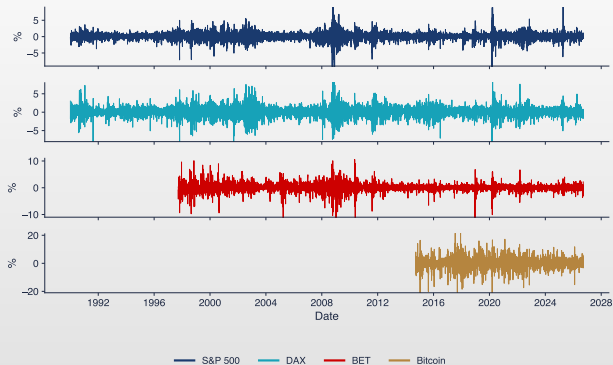
Amsterdam Stock Exchange, 21 October 1987, two days after the crash

Photo: [Bart Molendijk / Anefo \(1987\)](#); CC0; [Wikimedia Commons](#)

Volatility: Definition and Units

- Daily log return in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$ ([Chapter 1](#))
 - ▶ **volatility** = the standard deviation of returns, $\sigma = \sqrt{\text{Var}(r_t)}$
 - ▶ it has the units of the return (% per day); the variance σ^2 has units of %²
- **Annualisation**: with a observations per year and uncorrelated returns, $\sigma_{\text{year}} = \sigma_{\text{day}} \sqrt{a}$
 - ▶ a = the actual frequency: about 252 for the S&P 500, 250 for the BET, 365 for Bitcoin (trades every day)
 - ▶ the rule $\sqrt{a}$ is the square-root-of-time rule of [Chapter 7](#): it needs uncorrelated returns
- Volatility measures dispersion in both directions: a large gain counts as much as a large loss
 - ▶ other measures of uncertainty exist, for example the information entropy of returns [Pele, Lazar & Dufour \(2017\)](#)

Four Markets, One Pattern



- Daily log returns in %: S&P 500 and DAX since 1990, BET since 1997, Bitcoin since 2014; last day 18 September 2026
- Quiet years alternate with turbulent months in all four series

Interpretation: Volatility in Four Markets

	Annual volatility	Largest daily move	Date	Excess kurtosis
S&P 500	18.0%	−12.8%	16 March 2020	10.9
DAX	21.8%	−13.1%	12 March 2020	6.0
BET	23.4%	−13.1%	7 January 2009	9.9
Bitcoin	66.6%	−46.5%	12 March 2020	12.0

- Bitcoin is about 20% more volatile per year than its daily figure suggests if one wrongly uses $\sqrt{252}$: it trades 365 days a year
- The largest moves are 8 to 13 standard deviations: practically impossible under the Normal distribution ([Chapter 2](#))
- Excess kurtosis far above 0 (the Normal value): heavy tails; Section 6 shows that clustering is one cause

 [SFM_ch8_returns_volatility](#)

Volatility Is Latent

- Model: $r_t = \mu + \sigma_t z_t$, with z_t i.i.d., mean 0, variance 1
 - ▶ σ_t is the **conditional volatility**: $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$, with $\mathcal{F}_{t-1}$ the information at the close of day $t - 1$
 - ▶ the **unconditional volatility** $\sigma = \sqrt{\text{Var}(r_t)}$ is its long-run average level
- **Latent**: we never observe σ_t , only one return per day
 - ▶ r_t^2 is an unbiased but very noisy measure of σ_t^2 (with $\mu = 0$): $E[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$
 - ▶ a quiet day can happen in a turbulent period, and the other way round
- Three ways to get more information about σ_t
 - ▶ average many days (historical volatility, EWMA); use the whole daily path (high and low prices); use intraday prices (realised volatility)

Worked Example: Annualising Volatility

- ▣ S&P 500 since 1990: daily standard deviation $\hat{\sigma} = 1.13\%$, $a = 252$ observations per year
 - ▶ $\hat{\sigma}_{\text{year}} = 1.13 \times \sqrt{252} = 1.13 \times 15.87 = 18.0\%$
- ▣ Bitcoin since 2014: $\hat{\sigma} = 3.49\%$ per day, 7 days a week
 - ▶ correct: $3.49 \times \sqrt{365} = 3.49 \times 19.11 = 66.6\%$
 - ▶ wrong, with the equity convention: $3.49 \times \sqrt{252} = 55.3\%$
- ▣ Comparing assets: annualise each with its own calendar, then compare

Recap: What Is Volatility?

- Volatility = standard deviation of returns; annualise with $\sqrt{a}$, a the actual number of observations per year
- Conditional volatility σ_t changes over time and is never observed: we can only estimate it
- All markets show quiet and turbulent periods

Historical (Rolling) Volatility

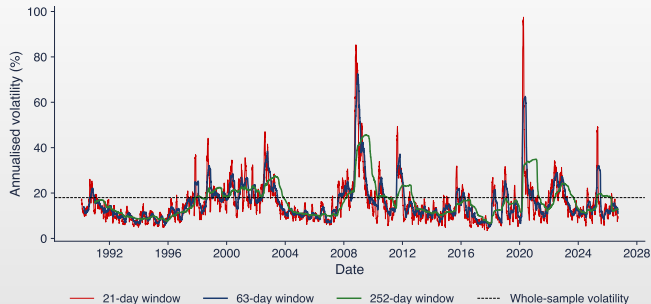
- **Historical volatility** on a window of the last n days:

- ▶ $\hat{\sigma}_t = \sqrt{\frac{1}{n-1} \sum_{j=0}^{n-1} (r_{t-j} - \bar{r}_t)^2}$, with $\bar{r}_t$ the mean of the same n returns
- ▶ **rolling window**: each day the newest return enters and the oldest leaves

- Zero-mean version: $\hat{\sigma}_t^2 = \frac{1}{n} \sum_{j=0}^{n-1} r_{t-j}^2$

- ▶ for daily data the mean is tiny compared with σ (0.03% against 1.13% for the S&P 500); setting it to 0 removes a noisy estimate
- Usual windows: $n = 21$ (a month), 63 (a quarter), 252 (a year) of trading days
- ▶ each $\hat{\sigma}_t$ is annualised with $\sqrt{a}$ and plotted at the last day of its window

S&P 500: Three Windows, Three Answers



- Rolling volatility of the S&P 500, annualised; dashed line: the whole-sample volatility, 18.0%
- Question for the room:** which window would you trust on the day after a crash?

Interpretation: the Window Trade-Off

- Short window ($n = 21$): reacts fast, but is noisy
 - ▶ maximum 97.5% on 6 April 2020; the typical daily change of $\ln \hat{\sigma}_t$ is 6.7%
- Long window ($n = 252$): smooth, but slow
 - ▶ maximum only 45.6%, reached on 15 July 2009, months after the peak of the 2008 crisis; daily change 0.8%
- **Answer:** a short window, or EWMA (next section), because a long window still averages the calm days before the crash
 - ▶ for capital planning, a long window is preferred: it changes slowly
- At the end of the sample: 9.6% (21 days), 11.1% (63 days), 12.9% (252 days)

How Precise Is a Sample Volatility?

- With n i.i.d. Normal returns, the relative standard error of $\hat{\sigma}$ is $SE(\hat{\sigma})/\sigma \approx 1/\sqrt{2n}$
 - ▶ $n = 21$: 15%; $n = 63$: 9%; $n = 252$: 4%
- With heavy tails (kurtosis κ): $SE(\hat{\sigma})/\sigma \approx \sqrt{(\kappa - 1)/(4n)}$ (delta method)
 - ▶ S&P 500: $\kappa = 13.9$, so $n = 21$: 39%; $n = 252$: 11%
 - ▶ for $\kappa = 3$ (Normal) the two formulas coincide
- **Worked example:** the last 21-day volatility of the S&P 500 is 9.6%
 - ▶ approximate 95% interval, Normal returns: [6.7%; 12.5%]; with the observed kurtosis: [2.2%; 17.0%]

Recap: Historical Volatility

- Rolling standard deviation on n days; daily data: the mean can be set to zero
- Short windows react fast but are noisy; long windows are smooth but slow
- A 21-day volatility is uncertain by $\pm 15\%$ or more: one decimal is already too much precision

1994: RiskMetrics Makes Volatility a Daily Number

- J.P. Morgan launched RiskMetrics in October 1994 and published its methods and data free of charge [J.P. Morgan/Reuters \(1996\)](#)
 - ▶ goal: one common yardstick for market risk across banks
 - ▶ volatilities and correlations of more than 480 series, updated every day
- The method: an exponentially weighted moving average (EWMA) of squared returns
 - ▶ one parameter, the **decay factor** λ : 0.94 for daily forecasts, 0.97 for monthly ones
 - ▶ for each series, λ minimises the root mean squared error of its variance forecasts; 0.94 is the weighted average over all series



The J.P. Morgan building at 23 Wall Street, New York

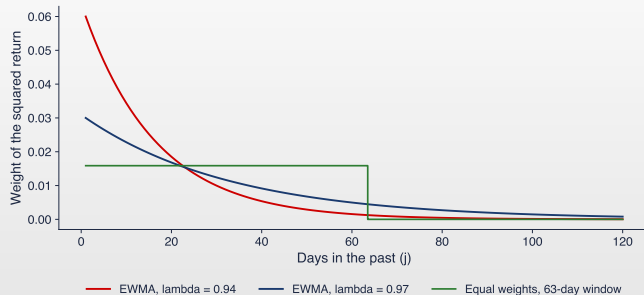
Photo: [Beyond My Ken \(2012\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

The EWMA Estimator

- **EWMA** recursion (zero mean): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, $0 < \lambda < 1$
 - ▶ σ_t^2 uses returns up to day $t - 1$: it is the forecast for day t
 - ▶ start: σ_1^2 = the mean of the first squared returns; its effect fades after a few months
- Unrolling the recursion: $\sigma_t^2 = (1 - \lambda) \sum_{j \geq 1} \lambda^{j-1} r_{t-j}^2$
 - ▶ weights $(1 - \lambda) \lambda^{j-1}$ decline geometrically and sum to 1
 - ▶ **half-life**: the lag at which the weight halves, $\ln 0.5 / \ln \lambda$
- A rolling window gives weight $1/n$ to each of the last n days and 0 to all older days

EWMA Weights against a Rolling Window

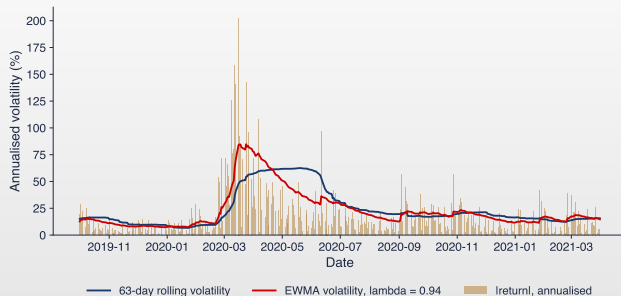


- $\lambda = 0.94$: half-life 11.2 days, mean lag $1/(1 - \lambda) = 16.7$ days, 99% of the weight in the last 74 days
- $\lambda = 0.97$: half-life 22.8 days, 99% of the weight in the last 151 days

Worked Example: One EWMA Update

- Yesterday's EWMA variance: $\sigma_{t-1}^2 = 1.00$ ($\%^2$), a daily volatility of 1%; yesterday's return: $r_{t-1} = -3\%$
 - ▶ step 1: $\lambda\sigma_{t-1}^2 = 0.94 \times 1.00 = 0.94$
 - ▶ step 2: $(1 - \lambda)r_{t-1}^2 = 0.06 \times 9 = 0.54$
 - ▶ step 3: $\sigma_t^2 = 0.94 + 0.54 = 1.48$, so $\sigma_t = 1.22\%$ per day, $1.22 \times \sqrt{252} = 19.3\%$ per year
- If today is quiet ($r_t = 0$): $\sigma_{t+1}^2 = 0.94 \times 1.48 = 1.39$
 - ▶ the shock fades by 6% per day; it never drops out abruptly
- A 63-day window would add $9/63 = 0.14$ to the variance today and remove this contribution at once, 63 days later

The Ghost Effect: March–June 2020



- S&P 500: 63-day rolling volatility and EWMA volatility ($\lambda = 0.94$), annualised; bars: $|r_t|$ on the same annual scale
- **Question for the room:** why does the blue line fall sharply in mid-June 2020, on a calm day?

Interpretation: Ghosts and Fading Memory

- **Answer:** on 15 June 2020 the return of 16 March 2020 (-12.8%) left the 63-day window
 - ▶ the rolling volatility fell from 50.4% to 43.0% in one day, with no news at all
 - ▶ this artefact is the **ghost effect**: an old shock leaves the window abruptly
- EWMA reacts faster and forgets smoothly
 - ▶ EWMA peak: 84.8% on 24 March 2020; 63-day peak: 62.6% only on 20 May 2020
 - ▶ on the ghost day EWMA changed by only -0.9 percentage points
- Price of the speed: EWMA is noisier than a long window and has no long-run level to return to

EWMA and GARCH: a Pointer to Chapter 9

- **ARCH** Engle (1982) and **GARCH** Bollerslev (1986): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ EWMA is the special case $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$, so $\alpha + \beta = 1$ ("integrated" GARCH, **IGARCH**)
- Consequences for forecasts made at day t for day $t + h$
 - ▶ EWMA: flat, $E_t[\sigma_{t+h}^2] = \sigma_{t+1}^2$ for every h : today's level persists for ever
 - ▶ GARCH with $\alpha + \beta < 1$: pulled towards the long-run variance $\omega/(1 - \alpha - \beta)$
- Chapter 9 estimates ω , α , β by maximum likelihood; here λ is fixed or chosen by a forecast criterion (Section 8)

Recap: EWMA and RiskMetrics

- ▣ $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$; RiskMetrics: $\lambda = 0.94$ for daily data
- ▣ Geometric weights: half-life 11.2 days for $\lambda = 0.94$; no ghost effect
- ▣ EWMA = IGARCH(1,1) without a constant: flat forecasts, no mean reversion

Open, High, Low and Close Prices

- **OHLC** data: for each day t , the open O_t , the high H_t , the low L_t and the close C_t
 - ▶ the close-to-close return uses 2 prices a day; H_t and L_t carry information about the whole path of the day
- Notation (log differences in %), each from the open of day t :
 - ▶ $u_t = \ln(H_t/O_t)$, $d_t = \ln(L_t/O_t)$, $c_t = \ln(C_t/O_t)$; so $d_t \leq 0 \leq u_t$ and $d_t \leq c_t \leq u_t$
 - ▶ **overnight** return $o_t = \ln(O_t/C_{t-1})$; the daily return is $r_t = o_t + c_t$
 - ▶ the **range** $u_t - d_t = \ln(H_t/L_t)$
- Data: S&P 500 since 2008, DAX since 2006, Bitcoin since 2014, BVB stocks since 2010; the BET file has closes only

One Trading Day: What OHLC Records



- Simulated log price (Brownian motion, 390 one-minute steps) after an overnight gap of 0.35%: the close keeps only the end point; high and low summarise the path
- Two days with the same close can have very different ranges, and so very different risk

The Parkinson Estimator (1980)

- For a Brownian motion with no drift and variance σ^2 per day: $E[(u_t - d_t)^2] = 4 \ln 2 \sigma^2$ Parkinson (1980)
 - ▶ one day: $\hat{\sigma}_{P,t}^2 = \frac{(u_t - d_t)^2}{4 \ln 2}$; on n days: the mean of these values
 - ▶ $4 \ln 2 \approx 2.77$: the range is on average larger than σ
- Assumptions
 - ▶ continuous trading during the day, so the observed high and low are the true extremes
 - ▶ no drift and no overnight jump: the range only sees the trading session
- Much more precise than r_t^2 : a simulation later in this section measures by how much
 - ▶ the log range is also used to estimate stochastic volatility models Alizadeh, Brandt & Diebold (2002)

Garman–Klass (1980) and Rogers–Satchell (1991)

- **Garman–Klass** Garman & Klass (1980): adds the open-to-close return
 - ▶ $\hat{\sigma}_{GK,t}^2 = 0.5(u_t - d_t)^2 - (2 \ln 2 - 1)c_t^2$
 - ▶ the weights make the estimator unbiased and nearly of minimum variance among such quadratic forms, under no drift
- **Rogers–Satchell** Rogers & Satchell (1991): unbiased for any drift μ
 - ▶ $\hat{\sigma}_{RS,t}^2 = u_t(u_t - c_t) + d_t(d_t - c_t)$
 - ▶ useful for assets with a strong trend over the window (for example, Bitcoin in a strong rally)
- Both still ignore the overnight return o_t : they measure the variance of the trading session only

Yang–Zhang (2000): Adding the Night

- On a window of n days Yang & Zhang (2000):

- ▶ $\hat{\sigma}_{YZ}^2 = \hat{\sigma}_O^2 + k \hat{\sigma}_C^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + (n + 1)/(n - 1)}$

- ▶ $\hat{\sigma}_O^2, \hat{\sigma}_C^2$: sample variances of the overnight returns o_t and of the open-to-close returns c_t ; $\hat{\sigma}_{RS}^2$: the mean Rogers–Satchell term

- Properties

- ▶ unbiased with drift and with opening jumps; k minimises the variance of the estimator
- ▶ $n = 21$: $k = 0.34/(1.34 + 22/20) = 0.34/2.44 = 0.14$

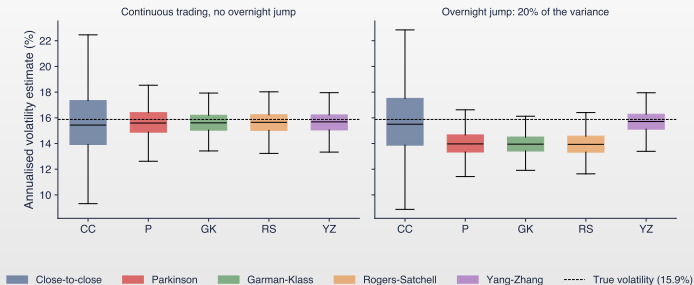
- Needs a window of $n \geq 2$ days (sample variances); Parkinson, Garman–Klass and Rogers–Satchell also work for a single day

Worked Example: S&P 500 on 16 March 2020

- ▣ Previous close 2711; open 2509, high 2563, low 2381, close 2386
 - ▶ $o = -7.76$, $u = 2.14$, $d = -5.22$, $c = -5.00$ (in %); close-to-close $r = o + c = -12.77$
- ▣ One-day variance estimates (%²)
 - ▶ close-to-close: $r^2 = 163.0$
 - ▶ Parkinson: $(7.37)^2/2.773 = 19.6$; Garman–Klass: 17.5; Rogers–Satchell: 16.5
 - ▶ overnight part $o^2 = 60.2$; overnight plus Rogers–Satchell: 76.7
- ▣ Most of the fall happened overnight, before the open: estimators that ignore o_t miss most of the risk of this day

 SFM_ch8_range_estimators

Efficiency by Simulation



- 1000 windows of 21 days; true volatility 1% per day (15.9% per year); intraday Brownian motion on 4680 steps (one every 5 seconds)
- Efficiency** of an estimator: $\text{Var}(\hat{\sigma}_{CC}^2)/\text{Var}(\hat{\sigma}^2)$, with CC the close-to-close estimator; right panel: 20% of the daily variance comes overnight

Interpretation: Precision Is Not Accuracy

	CC	Parkinson	GK	RS	YZ
Efficiency, continuous trading	1.0	5.4	9.3	7.6	8.5
Bias (%), continuous trading	0	-2	-3	-3	-2
Bias (%), overnight jump	1	-22	-22	-22	-2
MSE ratio, overnight jump	1.0	1.8	1.8	1.8	9.0
Bias (%), 26 prices a day	0	-23	-32	-32	-28

- With continuous trading, range estimators are 5 to 9 times more efficient: one month of ranges carries as much information as 5 to 9 months of closes
- With an overnight jump, Parkinson, GK and RS miss the night (bias about -22%); only YZ stays unbiased; **MSE** (mean squared error) ratio = $\text{MSE}(\text{CC})/\text{MSE}$
- With few trades the observed range is too small: all range estimators underestimate (thin markets) Molnár (2012)

Real Data: Five Estimators, Five Assets

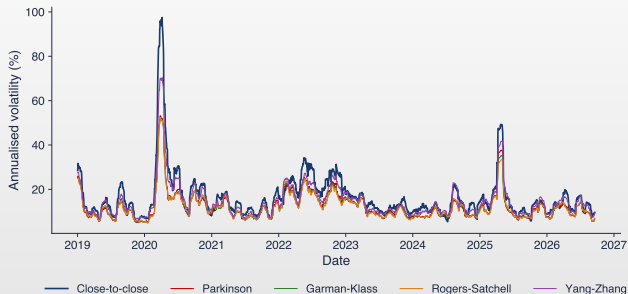
	Since	CC	P	GK	RS	YZ	Night share	corr(o, c)
S&P 500	2008	19.9	15.4	14.6	14.4	16.3	13%	0.22
DAX	2006	20.9	16.7	16.6	16.6	20.3	31%	0.02
Bitcoin	2014	66.6	63.5	62.4	62.5	63.2	0%	0.02
TLV	2010	26.6	21.5	20.6	21.0	25.6	26%	-0.08
SNP	2010	24.5	21.4	20.8	21.4	25.8	27%	-0.20

- Annualised volatility (%) on the whole OHLC sample; P = Parkinson; night share = $\hat{\sigma}_O^2 / (\hat{\sigma}_O^2 + \hat{\sigma}_C^2)$
- Stocks: open, high and low adjusted with the same factor as the close; days without trades removed

Interpretation: Why the Estimators Disagree

- Bitcoin trades 24 hours a day: the night share is almost 0, and all five estimators agree within a few points
 - ▶ the remaining gap: the daily path is observed in discrete trades, not continuously
- DAX and the BVB stocks: a quarter to a third of the variance comes overnight
 - ▶ Parkinson, GK and RS are far below CC; YZ, which adds the night, is close to CC
- S&P 500: even YZ (16.3%) stays below CC (19.9%)
 - ▶ overnight and open-to-close returns are positively correlated (0.22); a likely reason: the index open uses stale prices of stocks that have not traded yet
 - ▶ $\text{Var}(r) = \text{Var}(o) + \text{Var}(c) + 2 \text{Cov}(o, c)$: no range estimator sees the covariance term
- Lesson: check the assumptions on the data before trusting an efficiency gain

S&P 500: Rolling 21-Day Estimates since 2019



- Annualised 21-day volatility by the five estimators; means since 2019: CC 16.3%, Parkinson 12.5%, YZ 14.7%
- The range-based lines are smoother: the typical daily change of the log estimate is 4.3% for Parkinson and 4.2% for YZ, against 7.1% for CC

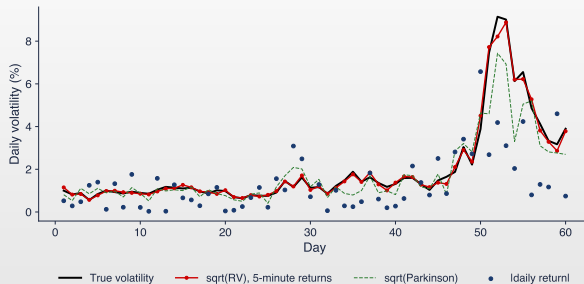
Recap: Range-Based Estimators

- ▣ Parkinson uses the range, Garman–Klass adds the open-to-close return, Rogers–Satchell allows a drift, Yang–Zhang adds the night
- ▣ In theory 5 to 9 times more efficient than close-to-close; in practice biased down by overnight jumps and discrete trading
- ▣ Choose Yang–Zhang for markets that close at night; any range estimator for 24-hour markets

Realised Variance from Intraday Returns

- Split day t into M intervals (for example $M = 78$ five-minute returns $r_{t,i}$ in a 6.5-hour session)
 - ▶ **realised variance** $RV_t = \sum_{i=1}^M r_{t,i}^2$; **realised volatility** $= \sqrt{RV_t}$
 - ▶ as $M \rightarrow \infty$, RV_t converges to the **integrated variance** of day t , the true variance of the day [Andersen, Bollerslev, Diebold & Labys \(2003\)](#), [Barndorff-Nielsen & Shephard \(2002\)](#)
- Volatility becomes almost observable: a nearly error-free measure for each day
 - ▶ used to evaluate forecasts (Section 8) and to forecast volatility directly, for example with the HAR model [Corsi \(2009\)](#)
 - ▶ high-frequency Bitcoin prices and daily risk: [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Our course data are daily; the next two slides use simulated one-minute prices to show how RV behaves

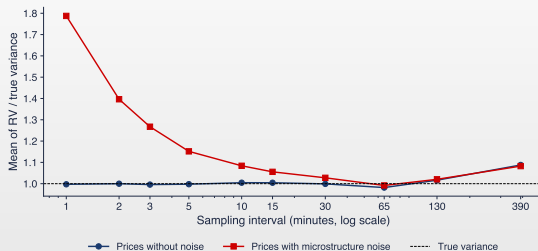
Simulated Example: RV Tracks the True Volatility



- 60 simulated days of 390 one-minute returns; the daily volatility follows a log AR(1) process (persistence 0.9)
- Mean absolute relative error: 5-minute RV 6%, Parkinson 23%, $|r_t|$ 58%; correlation with the truth: 0.995, 0.94 and 0.59

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Microstructure Noise and the Signature Plot



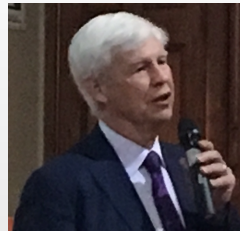
- Observed prices = true prices + **microstructure noise** (bid–ask bounce, price ticks), here with sd 0.025%; each squared return then adds $2\times$ the noise variance
- With noise, one-minute RV overstates the variance by a factor of 1.79; at 5 minutes: 1.15; the **signature plot** shows RV against the sampling interval
- In practice, 5-minute RV is hard to beat [Liu, Patton & Sheppard \(2015\)](#)

Recap: Realised Volatility

- ▣ $RV_t = \sum_i r_{t,i}^2$ estimates the variance of one day almost without error
- ▣ Too frequent sampling picks up microstructure noise; 5 minutes is the usual compromise
- ▣ Ranking of daily measures: $RV > \text{range estimators} > \text{squared daily return}$

1982: Engle and the ARCH Model

- **Volatility clustering:** periods of large absolute returns alternate with periods of small ones [Mandelbrot \(1963\)](#), [Cont \(2001\)](#)
 - ▶ returns are almost uncorrelated ([Chapter 7](#)), but $|r_t|$ and r_t^2 are strongly autocorrelated
- Robert F. Engle [Engle \(1982\)](#): the variance of today depends on yesterday's squared shocks: **ARCH**
 - ▶ the first model in which σ_t changes over time in a predictable way
 - ▶ Nobel Prize in Economic Sciences 2003, shared with Clive Granger [Nobel Prize \(2003\)](#); Engle's Nobel lecture: [Engle \(2004\)](#)
- This section: tests for clustering; [Chapter 9](#): the models

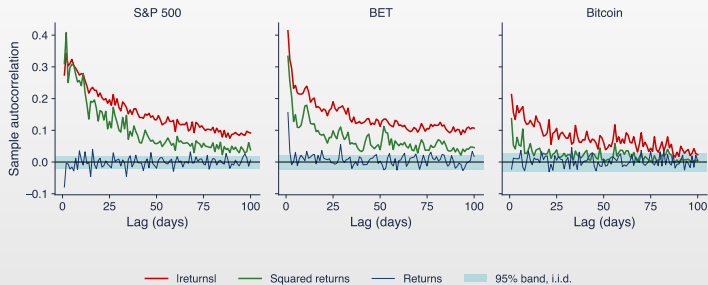


Robert F. Engle (b. 1942), Santiago, 2017

Photo: [Econterms \(2017\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

The Sign Is Unpredictable, the Size Is Not



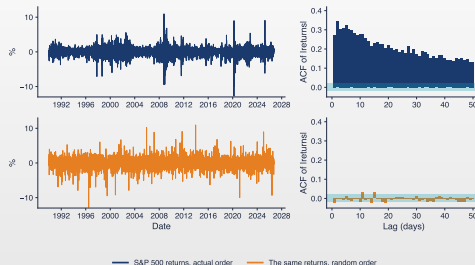
- Sample ACF (autocorrelation function, [Chapter 7](#)) of r_t , r_t^2 and $|r_t|$, lags 1–100, with the 95% band $\pm 1.96/\sqrt{T}$ for i.i.d. data
- S&P 500: all 100 autocorrelations of r_t^2 and of $|r_t|$ lie above the band; at lag 100, $|r_t|$ still has 0.09

Interpretation: Clustering in Three Markets

	$\hat{\rho}_1(r)$	$\hat{\rho}_{10}(r)$	$\hat{\rho}_{100}(r)$	$\hat{\rho}_1(r^2)$	$\hat{\rho}_{10}(r^2)$	Lags outside, $ r $
S&P 500	0.27	0.28	0.09	0.31	0.24	100/100
BET	0.41	0.23	0.11	0.33	0.18	100/100
Bitcoin	0.21	0.12	0.02	0.14	0.04	92/100

- Slow decay: a shock to volatility is still visible after months; [Chapter 11](#) calls this long memory
- **Taylor effect:** $|r_t|$ is more autocorrelated than r_t^2 (96% of lags for the S&P 500, all lags for the BET and Bitcoin) [Ding, Granger & Engle \(1993\)](#)
- Bitcoin: weaker clustering in r_t^2 (26 of 100 lags outside the band): a few huge days dominate the squares

Is Clustering Real? Shuffle the Days



- The same S&P 500 returns in a random order: same distribution and same kurtosis (excess 10.9), but no clustering
- $\hat{\rho}_1(|r|)$: 0.27 in the actual order, -0.02 after shuffling; Ljung–Box $Q(10)$ of r_t^2 : 8015 against 6.3
- Clustering is a property of the **order** of the returns, not of their distribution

Testing for Clustering: Ljung–Box on Squared Returns

- H_0 : no clustering, $\text{Var}(r_t | \mathcal{F}_{t-1})$ constant, so r_t^2 is uncorrelated
 - ▶ **McLeod–Li test** [McLeod & Li \(1983\)](#): the Ljung–Box statistic [Ljung & Box \(1978\)](#) applied to r_t^2
 - ▶ $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k(r^2)^2 / (T-k) \sim \chi^2(m)$ under H_0 ; reject if $Q(m)$ exceeds the 95% quantile
- **Worked example**: S&P 500, $T = 9\,240$, $m = 3$
 - ▶ $\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3$ of r_t^2 : 0.311, 0.408, 0.250
 - ▶ terms $T(T+2)\hat{\rho}_k^2 / (T-k)$: 892, 1 540, 579; $Q(3) = 3\,012 \gg \chi_{0.95}^2(3) = 7.81$: reject H_0
- The same test on r_t ([Chapter 7](#)) asks a different question: is the **mean** predictable?

The ARCH-LM Test (Engle, 1982)

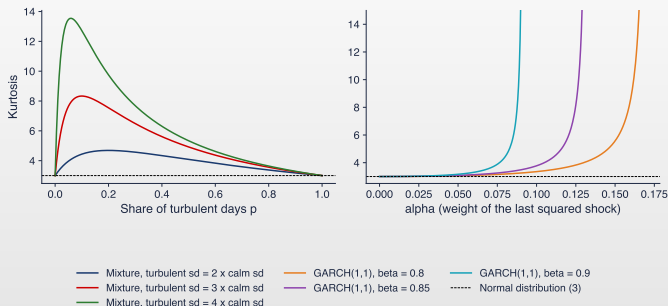
- ▣ Auxiliary regression of the squared demeaned returns e_t^2 , $e_t = r_t - \bar{r}$, on q of their own lags [Engle \(1982\)](#):
 - ▶ $e_t^2 = c_0 + c_1 e_{t-1}^2 + \dots + c_q e_{t-q}^2 + u_t$; $H_0: c_1 = \dots = c_q = 0$
 - ▶ **LM** (Lagrange multiplier) statistic: $LM = nR^2 \sim \chi^2(q)$, n = number of observations in the regression
- ▣ S&P 500, $q = 5$: $R^2 = 0.241$, $n = 9\,235$
 - ▶ $LM = 9\,235 \times 0.241 = 2\,227 \gg \chi_{0.95}^2(5) = 11.07$: strong ARCH effects
- ▣ Interpretation of R^2 : about a quarter of the day-to-day variation of e_t^2 is predictable from the last five days
- ▣ Both tests are diagnostics: they say that clustering exists, not how to model it ([Chapter 9](#))

Clustering Tests in Six Series

	T	$\hat{\rho}_1(r)$	$Q(10), r$	$\hat{\rho}_1(r^2)$	$Q(10), r^2$	$Q(10), r $	ARCH-LM(5)	Exc. kurt.
S&P 500	9 240	-0.079	90.8	0.311	8 015	8 323	2 227	10.9
DAX	9 288	-0.009	21.8	0.155	3 747	5 794	1 209	6.0
BET	7 243	0.156	198.3	0.333	2 500	5 184	1 057	9.9
Bitcoin	4 384	-0.022	20.2	0.137	213	1 086	115	12.0
TLV	3 788	0.010	29.1	0.122	481	1 179	253	13.5
SNP	3 815	-0.015	20.7	0.195	505	944	234	8.9

- 5% critical values: $\chi_{0.95}^2(10) = 18.31$ for $Q(10)$, $\chi_{0.95}^2(5) = 11.07$ for ARCH-LM(5); every statistic on r^2 and $|r|$ rejects with $p < 0.001$
- $Q(10)$ on r is far smaller: mean predictability is weak (Chapter 7), size predictability is strong everywhere
- Bitcoin and the two BVB stocks have the smallest ARCH-LM values despite heavy tails: a few extreme days dilute the regression

Clustering Creates Heavy Tails

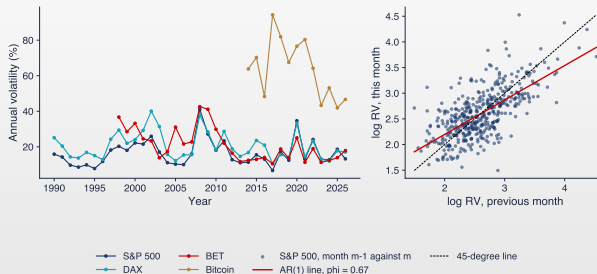


- Left: calm days with sd 1, turbulent days with sd 2 (share $p = 0.2$): $E[\sigma^2] = 1.6$, $E[\sigma^4] = 4.0$, kurtosis $3E[\sigma^4]/E[\sigma^2]^2 = 4.69$, though each day is Normal
- Right: GARCH(1,1) kurtosis $3 + 6\alpha^2/(1 - \beta^2 - 2\alpha\beta - 3\alpha^2)$, as in SFEkurgarch; $\alpha = 0.10$, $\beta = 0.85$: 3.77 (Chapter 9)

Recap: Volatility Clustering

- ▣ r_t almost uncorrelated, $|r_t|$ and r_t^2 strongly and persistently correlated: clustering
- ▣ Tests: Ljung–Box on r_t^2 (McLeod–Li) and ARCH-LM = nR^2 ; both reject in every market
- ▣ Clustering is about the order of returns, and it is one source of heavy tails

Volatility over Three Decades



- Left: volatility of each calendar year (annualised, %); right: log monthly realised volatility of the S&P 500 (from daily returns) against the previous month
- S&P 500: from 7% (2017) to 41% (2008); BET: 10% (2017) to 43% (2008); Bitcoin: 42% (2025) to 94% (2017)

Interpretation: Persistence and Mean Reversion

- ▣ Volatility changes over time, but it does not drift away: it returns to a normal level [Schwert \(1989\)](#)
 - ▶ AR(1) of log monthly realised volatility of the S&P 500 (441 months): $\hat{\phi} = 0.67$ (SE 0.04)
 - ▶ half-life of a volatility shock: $\ln 0.5 / \ln \hat{\phi} = 1.7$ months; long-run level about 13.5%
- ▣ Monthly volatility is right-skewed (skewness 3.3), its logarithm much less (0.6) [Andersen, Bollerslev, Diebold & Ebens \(2001\)](#)
 - ▶ mean 15.3% above the median 12.9%: a few crisis months pull the mean up
 - ▶ models of volatility are therefore often written for $\ln \sigma_t$
- ▣ Mean reversion is what EWMA lacks and GARCH with $\alpha + \beta < 1$ has ([Chapter 9](#))

 SFM_ch8_long_term_vix

The VIX: Implied Volatility

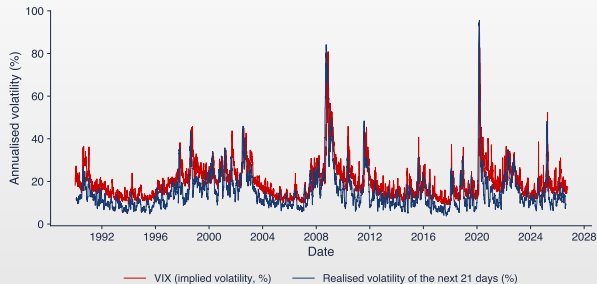
- Options trade on exchanges since 1973, when members of the Chicago Board of Trade founded the **CBOE** (Chicago Board Options Exchange)
 - ▶ an option price depends on the volatility the market expects until expiry
 - ▶ **implied volatility**: the volatility that, put into an option-pricing formula, gives the observed price
- **VIX**, introduced in 1993 [Whaley \(1993\)](#): the expected volatility of the S&P 500 over the next 30 calendar days, from S&P 500 option prices [Cboe \(2026\)](#)
 - ▶ quoted in annualised %: $VIX = 20$ means about $20/\sqrt{12} \approx 5.8\%$ over a month
 - ▶ called the “fear index”: it jumps when stocks fall [Whaley \(2009\)](#)



Traders at the Chicago Board of Trade, 31 May 1973

Photo: [U.S. Department of Agriculture \(1973\)](#); public domain; [Wikimedia Commons](#)

VIX against the Volatility That Followed



- VIX and the realised volatility of the S&P 500 over the next 21 trading days, $\sqrt{(a/21) \sum_{j=1}^{21} r_{t+j}^2}$; maximum VIX: 82.7 on 16 March 2020
- **Question for the room:** is the red line usually above or below the blue one?

Interpretation: Implied against Realised Volatility

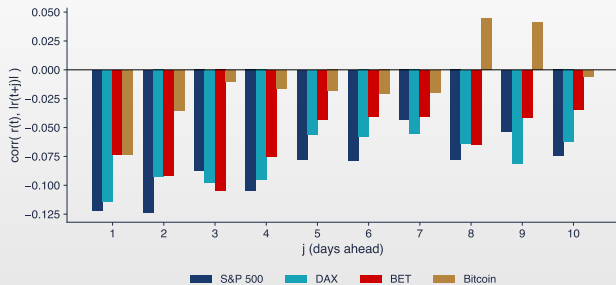
- **Answer:** above; VIX exceeded the next 21 days' realised volatility on 86% of the 9 199 days
 - ▶ mean VIX 19.4% against mean realised 15.3%: a gap of 4.1 points
 - ▶ **variance risk premium:** investors pay to be insured against volatility spikes Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- As a forecast, VIX is informative but biased
 - ▶ regression of future on implied: realised = $-1.72 + 0.88 \text{ VIX}$, $R^2 = 0.52$
 - ▶ the past 21-day realised volatility explains less: $R^2 = 0.44$
- Daily log changes of the VIX and S&P 500 returns: correlation -0.71 : fear rises when prices fall

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The Leverage Effect

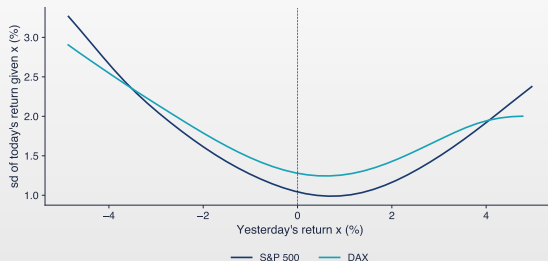
- **Leverage effect:** negative returns are followed by higher volatility than positive returns of the same size [Christie \(1982\)](#)
 - ▶ leverage story: a falling share price raises the debt-to-equity ratio, so the equity becomes riskier
 - ▶ volatility-feedback story: higher expected volatility raises the required return, so the price falls today [French, Schwert & Stambaugh \(1987\)](#), [Bekaert & Wu \(2000\)](#)
- A simple measure: $\text{corr}(r_t, |r_{t+j}|)$ for $j = 1, 2, \dots$
 - ▶ negative: today's fall announces tomorrow's turbulence; zero under a symmetric model
- [Chapter 2](#) met the asymmetry of the distribution; here the asymmetry is in the **dynamics** of volatility

Bad News Raises Future Volatility



- $\text{corr}(r_t, |r_{t+j}|)$, $j = 1, \dots, 10$; S&P 500: -0.122 at $j = 1$; the reverse direction $\text{corr}(|r_{t-1}|, r_t) = 0.025$
- Mean r_{t+1}^2 after a down day divided by that after an up day: S&P 500 1.51, DAX 1.46, BET 1.41, Bitcoin 1.06

Nonparametric Volatility Given Yesterday's Return



- $\hat{\sigma}(x) = \sqrt{\hat{m}_2(x) - \hat{m}_1(x)^2}$, $\hat{m}_1$, $\hat{m}_2$: local linear regressions of r_t and r_t^2 on $r_{t-1} = x$ (quartic kernel, bandwidth 0.04 on decimal returns, as in SFEvolnonparest)
- S&P 500: $\hat{\sigma}(-2\%) = 1.61\%$ against $\hat{\sigma}(+2\%) = 1.16\%$; the minimum is at $x \approx 0.7\%$, not at 0: an asymmetric news impact

Recap: Long-Run Behaviour, the VIX and the Leverage Effect

- ▣ Volatility is persistent but mean-reverting: half-life of about 1.7 months for the S&P 500
- ▣ VIX (implied) is above realised volatility on 86% of days: a variance risk premium
- ▣ Leverage effect: falls raise future volatility in equity indices; weaker and shorter-lived for Bitcoin

The Proxy Problem

- A forecast h_t of σ_t^2 is made at the close of day $t - 1$; the true σ_t^2 is never observed
 - ▶ we compare h_t with a **proxy** $\hat{\sigma}_t^2$: a measurable quantity with $E[\hat{\sigma}_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$
- Proxies, from noisy to precise: r_t^2 , a range estimator, realised variance
 - ▶ with r_t^2 even a perfect forecast looks poor: [Andersen & Bollerslev \(1998\)](#) showed that a low R^2 does not mean a bad model
- **Mincer–Zarnowitz regression** [Mincer & Zarnowitz \(1969\)](#): $\hat{\sigma}_t^2 = a + b h_t + u_t$
 - ▶ a good forecast has $a \approx 0$, $b \approx 1$; the R^2 measures how much of the proxy the forecast explains

Loss Functions: MSE and QLIKE

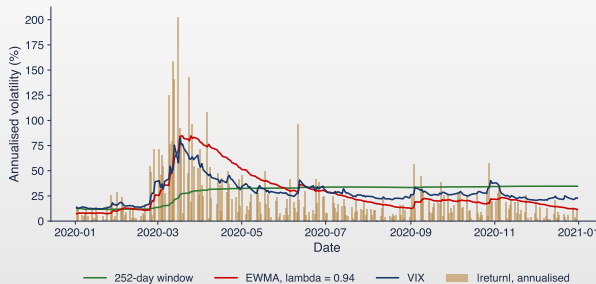
- Average loss over the evaluation days; lower is better [Patton \(2011\)](#)
 - ▶ MSE: $L = (\hat{\sigma}_t^2 - h_t)^2$
 - ▶ **QLIKE** (quasi-likelihood): $L = \hat{\sigma}_t^2/h_t + \ln h_t$, minus the Normal log-likelihood up to constants
- Both are **robust**: with a noisy but unbiased proxy they rank forecasts as the true variance would
 - ▶ losses on $\sqrt{h_t}$ or on $\ln h_t$ are not robust and can prefer the wrong forecast
- **Worked example**: true variance and proxy equal to 1; two forecasts, $h = 0.5$ and $h = 2$
 - ▶ MSE: $(1 - 0.5)^2 = 0.25$ against $(1 - 2)^2 = 1$: MSE punishes over-prediction more
 - ▶ QLIKE: $1/0.5 + (-0.693) = 1.307$ against $1/2 + 0.693 = 1.193$: QLIKE punishes under-prediction more, the costly error for risk management

S&P 500: Which Forecast Is Best?

	21 days	63 days	252 days	EWMA 0.94	EWMA 0.97	VIX
MSE	18.99	20.77	21.43	17.77	19.02	16.70
QLIKE	0.876	0.914	1.099	0.808	0.844	0.799
MZ R^2 , proxy r^2	0.14	0.05	0.01	0.17	0.11	0.25
MZ R^2 , range proxy	0.24	0.09	0.02	0.29	0.19	0.42

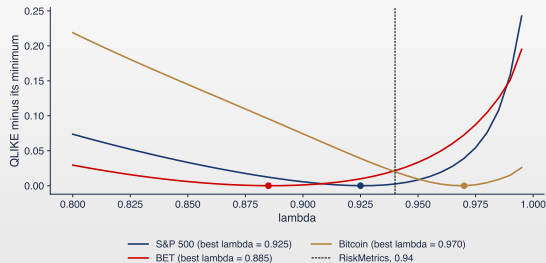
- Next-day variance forecasts for 4 202 days from 2010; proxy r_t^2 ; VIX forecast: VIX^2/a ; range proxy: σ_t^2 plus the Garman–Klass term
- **DM** (Diebold–Mariano) test of equal QLIKE [Diebold & Mariano \(1995\)](#): EWMA 0.94 against 21 days $t = -4.50$ ($p < 0.001$); VIX against EWMA 0.94 $t = -0.32$ ($p = 0.751$)
- EWMA beats the 21-day window; VIX and EWMA are not significantly different; the long window is worst; a precise proxy raises every R^2 , but not the ranking

Forecasts in 2020



- Next-day volatility forecasts of the S&P 500 in 2020 (annualised): 252-day window, EWMA ($\lambda = 0.94$) and VIX; bars: $|r_t|$
- VIX started to rise one trading day before the first large fall of February 2020; EWMA rose right after it; the 252-day window moved slowly and stayed near its peak for the rest of the year

Choosing λ by QLIKE



- Mean QLIKE of next-day EWMA forecasts (proxy r_t^2 , from 2010; Bitcoin from 2016) minus its minimum, for λ from 0.80 to 0.995
- Best λ : S&P 500 0.925, BET 0.885, Bitcoin 0.970; the RiskMetrics 0.94 loses only 0.002 for the S&P 500
- One λ for all assets is a convenient compromise, not an optimum

Recap: Evaluating Volatility Forecasts

- Compare forecasts with a proxy; r_t^2 is unbiased but noisy, so R^2 values are low even for good forecasts
- Use robust losses (MSE, QLIKE) and test the differences (Diebold–Mariano); a large comparison of volatility models: [Hansen & Lunde \(2005\)](#)
- S&P 500: EWMA and VIX are the best simple forecasts; long windows are the worst

An Open Question

- **Do range-based estimators help on the Bucharest Stock Exchange, where many stocks trade thinly?**
 - ▶ theory: ranges are 5 to 9 times more efficient; but thin trading shrinks the observed range (Section 4)
 - ▶ days with trades but an unchanged close, since 2010: TLV 9%, SNP 8%, BRD 10%, TGN 10%
 - ▶ Parkinson / close-to-close volatility: TLV 0.81, SNP 0.87, BRD 0.90, TGN 0.87 (S&P 500: 0.78)
 - ▶ QLIKE of next-day forecasts by the 21-day window, CC against YZ: TLV 2.022 against 2.039; TGN 1.750 against 1.636
- Open: the answer differs by stock; is it liquidity, the opening auction, or the size of the overnight move?
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

How AI Could Help

- ▣ **Literature:** list studies of range-based and realised volatility in emerging and thin markets, with their data and estimators
- ▣ **Code:** a first draft of the five estimators, rolling forecasts and QLIKE comparisons for all BET stocks
- ▣ **Robustness:** liquidity groups by volume, other windows, the 2016–2020 and 2021–2026 halves
- ▣ Example prompt
 - ▶ Write a Python function that computes the Parkinson, Garman-Klass, Rogers-Satchell and Yang-Zhang variance on rolling 21-day windows from daily open, high, low and close prices, and compares their next-day forecasts with QLIKE against squared close-to-close returns.

What to Check

- ▣ The constants: $4 \ln 2$ in Parkinson, $2 \ln 2 - 1$ in Garman–Klass, k in Yang–Zhang; a draft with $\ln 2$ or $k = 0.34$ is wrong
- ▣ Stock prices: open, high and low must be adjusted for dividends and splits with the same factor as the close
- ▣ Annualisation with the actual frequency: 365 for crypto, about 250 for BVB stocks
- ▣ No look-ahead: the forecast for day t uses only data up to day $t - 1$
- ▣ References: every cited paper must exist; check the DOI

Project Idea

- **Question:** which daily volatility estimator should a Romanian risk manager use?
 - ▶ data: BVB stocks since 2010 (TLV, SNP, BRD, TGN, and others with OHLC data), S&P 500 and DAX as benchmarks (EODHD)
- **Steps**
 - ▶ the five estimators on 21-day windows; EWMA versions of each; next-day forecasts
 - ▶ QLIKE and MSE with the proxy r_t^2 ; Diebold–Mariano tests; results by liquidity group
 - ▶ relate the gain of YZ over CC to the share of zero-volume days and to the overnight share
- **Deliverables:** one table, one chart, and a paragraph on what the data can and cannot show
- **Declare any AI use, and list the errors of the AI that you corrected**

Key Takeaways

- ▣ Volatility is the standard deviation of returns; it is latent and changes over time; annualise with the actual frequency
- ▣ Historical windows trade speed against noise; EWMA ($\lambda = 0.94$) reacts fast and avoids ghosts
- ▣ Range estimators use the daily path: far more efficient, but biased by overnight jumps and thin trading; Yang–Zhang adds the night
- ▣ Clustering: $|r_t|$ and r_t^2 are autocorrelated; detect it with Ljung–Box on r_t^2 and ARCH-LM
- ▣ Volatility mean-reverts; VIX usually exceeds realised volatility; falls raise future volatility
- ▣ Compare forecasts with robust losses (QLIKE) and a proxy; test the differences

Key Formulas

Quantity	Formula
Annualised volatility	$\sigma_{\text{year}} = \sigma_{\text{day}} \sqrt{a}$
EWMA	$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \text{half-life } \ln 0.5 / \ln \lambda$
Parkinson	$(u_t - d_t)^2 / (4 \ln 2)$
Garman–Klass	$0.5(u_t - d_t)^2 - (2 \ln 2 - 1) c_t^2$
Rogers–Satchell	$u_t(u_t - c_t) + d_t(d_t - c_t)$
Yang–Zhang	$\hat{\sigma}_O^2 + k \hat{\sigma}_C^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = 0.34 / (1.34 + (n + 1) / (n - 1))$
Realised variance	$RV_t = \sum_i r_{t,i}^2$
McLeod–Li	$Q(m) = T(T + 2) \sum_{k=1}^m \hat{\rho}_k(r^2)^2 / (T - k) \sim \chi^2(m)$
ARCH-LM	$LM = nR^2 \sim \chi^2(q)$
MSE, QLIKE	$(\hat{\sigma}_t^2 - h_t)^2, \quad \hat{\sigma}_t^2 / h_t + \ln h_t$

Check Yourself

- **Question:** a stock has $H = 102$, $L = 98$ on one day; what is the Parkinson daily volatility?
 - ▶ **Answer:** $\ln(102/98) = 4.00\%$; $\sqrt{4.00^2/2.77} = 2.40\%$
- **Question:** Ljung–Box on returns does not reject, on squared returns it rejects strongly; what do you conclude?
 - ▶ **Answer:** the mean is unpredictable, the variance is predictable: volatility clustering without autocorrelation
- **Question:** why can EWMA with $\lambda = 0.94$ not forecast a return of volatility to its normal level?
 - ▶ **Answer:** $\alpha + \beta = 1$ and $\omega = 0$: its forecasts are flat at every horizon
- Next: [Chapter 9](#), ARCH and GARCH models

References (1/3)

- ▣ Alizadeh, S., Brandt, M.W., Diebold, F.X. (2002). Range-based estimation of stochastic volatility models. *Journal of Finance*, 57(3), 1047–1091.
- ▣ Andersen, T.G., Bollerslev, T. (1998). Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society, Series B*, 64(2), 253–280.
- ▣ Bekaert, G., Wu, G. (2000). Asymmetric volatility and risk in equity markets. *Review of Financial Studies*, 13(1), 1–42.
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T., Tauchen, G., Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11), 4463–4492.
- ▣ Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- ▣ Carr, P., Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*, 22(3), 1311–1341.
- ▣ Cboe Global Markets (2026). Cboe Volatility Index (VIX). Product page and methodology.
- ▣ Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- ▣ Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.

References (2/3)

- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Engle, R.F. (2004). Risk and volatility: econometric models and financial practice. *American Economic Review*, 94(3), 405–420.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- French, K.R., Schwert, G.W., Stambaugh, R.F. (1987). Expected stock returns and volatility. *Journal of Financial Economics*, 19(1), 3–29.
- Garman, M.B., Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1), 67–78.
- Hansen, P.R., Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889.
- J.P. Morgan/Reuters (1996). *RiskMetrics – Technical Document*, 4th ed. New York: Morgan Guaranty Trust Company.
- Liu, L.Y., Patton, A.J., Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1), 293–311.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- McLeod, A.I., Li, W.K. (1983). Diagnostic checking ARMA time series models using squared-residual autocorrelations. *Journal of Time Series Analysis*, 4(4), 269–273.
- Mincer, J.A., Zarnowitz, V. (1969). The evaluation of economic forecasts. In J.A. Mincer (ed.), *Economic Forecasts and Expectations*, 3–46. NBER.
- Molnár, P. (2012). Properties of range-based volatility estimators. *International Review of Financial Analysis*, 23, 20–29.
- Nobel Prize Outreach (2003). The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2003: Robert F. Engle III and Clive W.J. Granger. nobelprize.org.
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.

References (3/3)

- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily Value at Risk. *Entropy*, 21(2), 102.
- ▣ Poon, S.-H., Granger, C.W.J. (2003). Forecasting volatility in financial markets: a review. *Journal of Economic Literature*, 41(2), 478–539.
- ▣ Rogers, L.C.G., Satchell, S.E. (1991). Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4), 504–512.
- ▣ Schwert, G.W. (1989). Why does stock market volatility change over time? *Journal of Finance*, 44(5), 1115–1153.
- ▣ Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- ▣ Wang, H., Fu, T., Du, Y., Gao, W., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620, 47–60.
- ▣ Whaley, R.E. (1993). Derivatives on market volatility. *Journal of Derivatives*, 1(1), 71–84.
- ▣ Whaley, R.E. (2009). Understanding the VIX. *Journal of Portfolio Management*, 35(3), 98–105.
- ▣ Yang, D., Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3), 477–492.